Contrib. Plasma Phys. **56**, No. 6-8, 784 – 789 (2016) / **DOI** 10.1002/ctpp.201610062

Numerical Simulation Study of Plasma Flow in the GAMMA10/PDX End-cell Using a Fluid Code

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Received 08 September 2015, revised 21 January 2016, accepted 22 January 2016
Published online 08 July 2016

Key words GAMMA 10/PDX, multi-fluid code, plasma detachment, impurity injection.

In order to understand the effect of impurity injection into divertor plasma, a multi-fluid code was developed and applied to the region from west plug/barrier-cell to end cell in GAMMA 10/PDX. In this research, Ar is applied as impurity neutrals. To investigate the effect of impurity injection on plasma parameters, interactions between Ar impurity and hydrogen bulk plasma were included in this model. Under the condition of injecting neutral hydrogen and impurity Ar, the charge exchange loss between hydrogen ion and neutrals reaches to about 95% and the radiation loss for electron is enhanced to about 45%. Therefore, hydrogen ion and electron temperatures decrease from about 55 eV to 4 eV and 15 eV to 4 eV, respectively. The heat load on the target plate is also reduced by nearly 80% compared with the condition of low injection rate of neutral hydrogen and impurity. It is also observed that the particle flux increases with neutral density and then shows a tendency to saturate in the higher neutral density ($> 2.1 \times 10^{18}$ m^{-3}).

1 Introduction

In toroidal fusion devices, reduction of heat load on to the target plate is one of the most important issues. The divertor is exposed to high heat load. For concentrated heat flux in divertor region, erosion and sputtering are produced on the divertor plate. Therefore, it is necessary to reduce the heat load. GAMMA 10/PDX ("GAMMA 10" and Potential-control and Divertor-simulation eXperiments") is the world's largest linear device which is 27 m in length and utilizes many plasma heating devices with the same scale of the present day fusion devices, such as Electron Cyclotron Heating (ECH), Ion Cyclotron Range of Frequency (ICRF), and Neutral Beam Injection (NBI) systems [1-2]. In the central-cell high temperature plasma can be generated by applying ICRF and ECH [3].The end loss plasma which escapes from central-cell transports through the anchor-cell, subsequently to plug/barrier- cell and finally reaches the end-cell. The end loss flux of the GAMMA 10/PDX has been focused of attention because of its applicability and extensibility for studies that require high performance plasma flux such as plasma-divertor interaction simulation experiments [4]. In GAMMA 10/PDX, divertor simulation experiments have been started by using a divertor simulation experimental module (D-module) [5]. In the D-module, a V-shaped target made of tungsten has been installed. Injecting neutral or impurity gases in the end-cell of GAMMA 10/PDX is useful to reduce particle and heat load toward the divertor target. However, it is concerned that injection of neutral and/or impurity lead to the degradation of core plasma performance. It is important to investigate transport of impurity particles, together with the effect of heat reduction by impurity. Numerical simulation is useful for analyzing these phenomena. The combination use of the experimental results and the numerical simulation results is performed in the world by using linear devices [6-8]. In order to investigate the energy loss processes and the plasma behaviors under the same conditions of the divertor simulation experiments in the GAMMA 10/PDX, a simplified code for the end-cell has been developed [9, 10] based on the same fluid model as the B2-code, which has been originally developed by Braams for the numerical simulation of tokamak SOL and divertor plasmas [11]. In the present model, we have developed a 2D model of the mirror magnetic configuration in the GAMMA 10/PDX in the approximately same manner as the B2 code. The impurity plays a

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key role in divertors. For more details analysis of impurity injection, the single fluid code has been extended to a multi-fluid model.

2 The mesh structure

In GAMMA 10/PDX, the plasma flows from the central-cell through the plug/barrier cell to the end-cell. On the other hand, recycling hydrogen neutrals, injected neutral hydrogen and impurity gas in the end-cell are transported toward the central region through the plug/barrier-cell. In this research, the numerical meshes are created on the basis of the GAMMA 10/PDX magnetic field configuration. In our present model, a 2D model parallel and perpendicular to the magnetic field is employed under the assumption of axial symmetry. Therefore, the local curvilinear orthogonal coordinate system (x, y) has been employed instead of the usual cylindrical coordinate system (r, z). The x-direction and y-direction in the local orthogonal coordinate system are corresponding to the direction parallel and perpendicular to the magnetic field respectively. The present model assumes the axial symmetry and 2D model in the x and y direction, while physical quantities are uniform in the theta (θ) direction. Figure 1 shows the numerical mesh model and the magnetic field structure of GAMMA 10/PDX. The mesh structure has been constructed in such a way that each cell becomes a rectangle shape with its sides being locally orthogonal to each other. In addition, the tungsten target has been designed at the end of the mesh structure. The mesh structure is composed of quadrangular cells with $0 \sim 15$ cm in the radial (r axis) direction and 750 $\sim$1070 cm in the z axial direction. In Fig. 1(b), the numerical mesh for the domain is shown. Initial conditions are chosen according to the typical experimental results of GAMMA 10/PDX. The plasma parameters at the plasma upstream region are defined as fixed boundary (Dirichlet condition). We apply divertor boundary condition on the target plate. In the divertor boundary condition, the energy balance equations of ion and electron is defined by heat flux flowed into the target plate. For ion velocity along with magnetic field line, Bohm condition near the sheath entrance has been adopted.

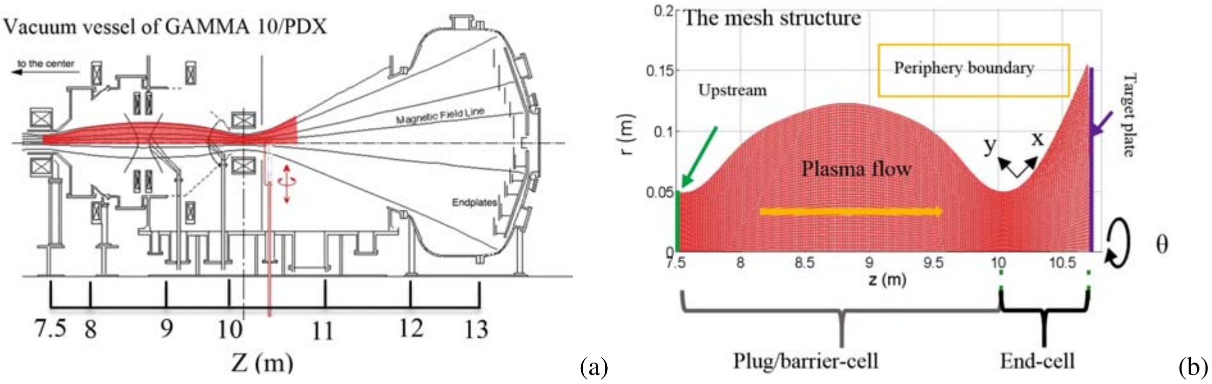

Fig. 1 Schemetic view of the GAMMA 10 west end-cell (a), and mesh structure in the simulation area (b).

3 Physical model

The fluid-code consists of five equations such as continuity equation, momentum equation, diffusion equation, ion energy-balance equation and electron energy-balance equation. Basic equations are discretized and numerically solved almost in the same way as in ref. [11].
Momentum balance equation of species α;

$$\frac{\partial}{\partial t}(m_\alpha n_\alpha u_{||\alpha}) + \frac{\partial}{\partial x}(m_\alpha n_\alpha u_\alpha u_{||\alpha} - \eta_x^a \frac{\partial u_{||\alpha}}{\partial x}) + \frac{\partial}{\partial y}((m_\alpha n_\alpha v_\alpha u_{||\alpha} - \eta_y^a \frac{\partial u_{||\alpha}}{\partial y}) = \frac{\mathcal{B}_\theta}{\mathcal{B}}[-\frac{\partial p_\alpha}{\partial x} - \frac{z_\alpha n_\alpha}{n_e}\frac{\partial p_e}{\partial x} + c_e(\frac{z_\alpha}{z_{eff}} - 1)Z_\alpha n_e \frac{\partial T_e}{\partial x} + c_i(\frac{z_\alpha}{z_{eff}} - 1)Z_\alpha n_\alpha \frac{\partial T_i}{\partial x}] + \sum_{\beta-1}^{N} F_{\alpha\beta} + S_{M||}^\alpha. \quad (1)$$

where, the symbols c_e and c_i are the coefficients of thermal force [12].
Multi-fluid model including the impurity ions is used in the following analyses. Impurity ions with a different charge state are treated as a different ion species. The ion energy balance equation for impurity ions is not solved in this model. Here, the impurity ion is added in ion energy balance equation as a term which is the work done by friction force between different ions species. In addition, the term which represents the energy exchange between impurity ion and electron by collision is introduced in the electron energy balance equation. By including these effects, ion and electron energy equations have been modified in the same way as in ref. [11]. The new electron and ion energy balance equation are as follows, respectively;

$$\frac{\partial}{\partial t}(\frac{3}{2}n_eT_e)+\frac{\partial}{\partial x}(\frac{5}{2}n_eu_eT_e-\kappa_x^e\frac{\partial T_e}{\partial x})+\frac{\partial}{\partial y}(\frac{5}{2}n_ev_eT_e-\kappa_y^e\frac{\partial T_e}{\partial y})=u_e\frac{\partial p_e}{\partial x}+v_e\frac{\partial p_e}{\partial y}-k(T_e-T_i)+S_E^e-k_z(T_e-T_z). \quad (2)$$

$$\frac{\partial}{\partial t}(\frac{3}{2}n_iT_i+\sum_\alpha\frac{1}{2}\rho_\alpha u_{||\alpha}^2)+\frac{\partial}{\partial x}[(\sum_\alpha\frac{5}{2}n_\alpha u_\alpha T_i+\sum_\alpha\frac{1}{2}m_\alpha n_\alpha u_\alpha u_{||\alpha}^2)-(\kappa_x^i\frac{\partial T_i}{\partial x}+\sum_\alpha\frac{1}{2}\eta_x^a\frac{\partial u_{||\alpha}^2}{\partial x})]$$
$$+\frac{\partial}{\partial y}[(\sum_\alpha\frac{5}{2}n_\alpha v_\alpha T_i+\sum_\alpha\frac{1}{2}m_\alpha n_\alpha v_\alpha u_{||\alpha}^2)-(\kappa_y^i\frac{\partial T_i}{\partial y}+\sum_\alpha\frac{1}{2}\eta_y^\alpha\frac{\partial u_{||\alpha}^2}{\partial y})]$$
$$=-u_e\frac{\partial p_e}{\partial x}-v_e\frac{\partial p_e}{\partial y}+k(T_e-T_i)+S_E^i+\sum_{\alpha\beta}F_{\alpha\beta}(u_{||\beta}-u_{||\alpha}). \quad (3)$$

Here the following notations are used: n_e and n_i are the density of electrons and ions; m_α and Z_α are the mass and charge number of species α; v_e and v_i are the average (drift) velocity of electron and ion; T_e and T_i are the electron and ion temperatures; p_e is the pressure of electrons; u is the flow velocity; B and B_θ are the magnetic induction and the component of magnetic field; $F_{\alpha\beta}$ is the friction force between species α and β; Z_{eff} is the effective charge; $S_{M||}^\alpha$, S_E^e , S_E^i are the parallel momentum, electron and ion energy due to interaction with neutral; η and κ are the viscosity coefficient and thermal conductivity; k represents the equipartition energy transfer.

As for the parallel transport, classical transport coefficients are used. The transport coefficients are different between single fluid and multi fluid. In the case of single fluid, the transport coefficients are indicated in ref. [12]. On the other hand, the coefficients used in multi-fluid are defined as simplification of the complete multi-species transport theory [13]. For neutral hydrogen injection, it is assumed that the injected neutral hydrogen distributes uniformly in the end region (10.055 m $\leq z \leq 10.705$ m) and decrease exponentially in the plug/barrier region (7.5 m $\leq z \leq 10.045$ m). At the present simulation conditions the following equations are assumed to describe hydrogen neutral density distribution. In order to minimize the effect of neutral injection in the plug/barrier region, a very small decay length constant (l_d) has been assumed.

$$n_o(z)=n_o\,exp(1-\int_{7.5}^{10.045}(\frac{1}{\lambda})dz)+n_{add}\,exp(-(\frac{\delta z}{l_d})^2), \quad (4)$$

$$n_o(z)=n_o\,exp(1-\int_{10.055}^{10.705}(\frac{1}{\lambda})dz)+n_{add}, \quad (5)$$

where, $\lambda=u_{ao}/(n_e<\sigma v>_{ionization}+n_e<\sigma v>_{CX})$ and $\delta z=z(i,j)-z(10.055,j)$.
The following notations are used: n_o is the background plasma density; n_{add} is the injecting hydrogen neutral density; $<\sigma v>$ is the reaction rate, u_{ao} is the particle velocity, i and j represent the number of mesh in the y and x axis, respectively. In the case of Ar injection, it is assumed that neutral Ar distributes uniformly from plug/barrier region to end region.
In the GAMMA 10/PDX, ion temperature is considerably high, therefore, charge-exchange (CX) reaction is thought to be a dominant process. Atomic processes in this code are shown in Table 1.

Table 1 Atomic processes are considered in this code [14]

Neutral gas	Ionization	Recombination	CX	Radiation loss
Hydrogen	H^{+}+ e → H^{+} + e + e	H^{+} + e→ H	H^{+} + H → H + H^{+}, and H^{+} + Ar → H + Ar^{+}	
Argon	Ar + e →Ar^{+} + e + e	Ar^{+} + e →Ar	H^{+} + Ar → H + Ar^{+}	Ar* + e →Ar + e +hν

4 Simulation results and discussion

An example of simulation results are shown in Fig. 2. Two peaks of electron density has been observed in the Fig. 2(a). It is natural to form the distribution like Fig. 2(a) because the magnetic configuration of the plug/barrier cell is mirror magnetic field. The peak positions of electron density correspond to the peak of mirror magnetic field. The distributions of ion and electron temperature are shown in Fig. 2(b) and (c), respectively. In this study, the Ar neutral density is fixed at 4.0 $\times$ 10^{17} m^{-3} , while the neutral hydrogen density is changed from 5.0 $\times$ 10^{17} m^{-3} to 2.3 $\times$ 10^{18} m^{-3}. In addition, higher Ar neutral density with 8.0 $\times$ 10^{17} m^{-3} is examined at neutral hydrogen densities of 5.0 $\times$ 10^{17}, 1.3$\times$ 10^{18} and 2.1 $\times$ 10^{18} m^{-3}. The simulated results are summarized in Figs. 3 and 4. The ion temperature and hydrogen ion density are shown as a function of the injected hydrogen density in Fig. 3 (a). The dependence of the electron temperature on the injected hydrogen density is shown in Fig. 3(b). At lower hydrogen gas density region ($<$ 1.9$\times$ 10^{18} m^{-3}), H^{+} temperature decreases almost linearly. On the other hand, at higher hydrogen density region ($>$ 1.9$\times$ 10^{18} m^{-3}), reduction of ion temperature becomes low. In the cases of Ar: 4.0 $\times$ 10^{17} and 8.0 $\times 10^{17}$ m^{-3} injection, H^{+} temperature and density is almost same as shown in Fig. 3(a).

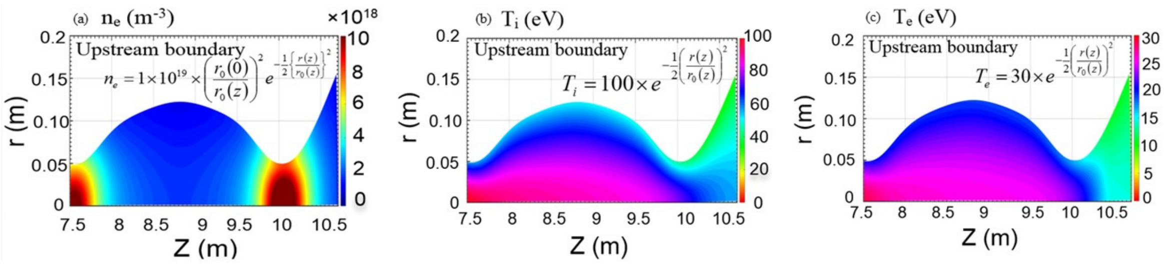

Fig. 2 Two-dimensional distributions of the simulation results (a): electron density n_e, (b): ion temperature T_i, and (c): electron temperature T_e

For Ar : 4.0 $\times$ 10^{17} m^{-3}, H^{+} density rises from 1.0 $\times$ 10^{18} m^{-3} to 3.3 $\times$ 10^{19} m^{-3} with the increment of injected neutral hydrogen density. As shown in Fig. 3 (b), electron temperature decreases from 16 eV to 4 eV with the increment of injected neutral hydrogen density. However, the electron temperature is falling regardless of low or high injected neutral hydrogen range. The charge-exchange (CX) reaction and the radiation loss for electron are strongly enhanced under the relatively high density range of the injected hydrogen ($>$ 1.9$\times$ 10^{18} m^{-3}). In Fig. 4, the dependence of the heat and particle fluxes on injected hydrogen density is shown. The heat load on the target plate reduces with the injected hydrogen neutral density. According to the increase of the hydrogen neutral density, the heat flux decreases. The heat flux reduces from 13 MW/m^2 to 3 MW/m^2 with varying the hydrogen neutral density from 5.0 $\times$ 10^{17} m^{-3} to 2.3 $\times$ 10^{18} m^{-3}. Particle flux at the target plate increases about 10 times when the injected neutral hydrogen density rises from 5 $\times$ 10^{17} m^{-3} to 2.3 $\times$ 10^{18} m^{-3}. It seems that the end-loss plasma approaches to the plasma detachment state because the particle flux goes to saturate at higher injected neutral hydrogen density ($>$ 2.1$\times$ 10^{18} m^{-3}).

The interaction of charged and neutral particles for ions or electron includes several processes. For H^{+} energy loss term, ionization, CX loss between hydrogen ion and neutrals, recombination and CX loss between hydrogen ion and Ar neutrals are considered. For electrons, ionization, recombination for H^{+} and Ar^{+} and inelastic collision with Ar are considered. Table 2 shows the rate of atomic molecular interaction for ion and electron. For ions, it is found that the most effective process is CX loss between hydrogen ion and neutral particles. The CX loss between hydrogen ions and neutrals reaches to about 96 %. Therefore, other processes such as recombination for H^+ and CX loss with Ar are very little existent. For electrons, on the other hand, the recombination of H^+ or Ar^+ is hardly any influence for electron energy loss. The effective processes are the effect of the ionization for H and inelastic collision with neutral Ar particles. Especially, the radiation loss by inelastic collision with neutral Ar keeps to be large influence despite the amount of neutral Ar injected into plasma which is lower than the injected neutral hydrogen. Therefore, the effects of radiation loss by injecting impurity gas are important to achieve the reduction of electron energy.

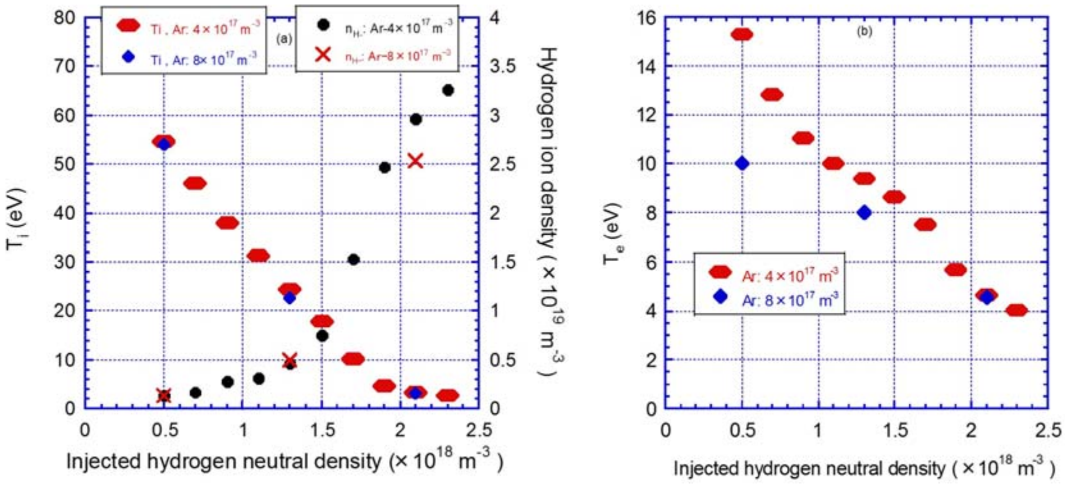

Fig. 3 The plasma parameters of the target plate at the z-axis dependence on injected neutral hydrogen density. (a): Ion temperatures and H+ densities and (b): Electron temperatures.

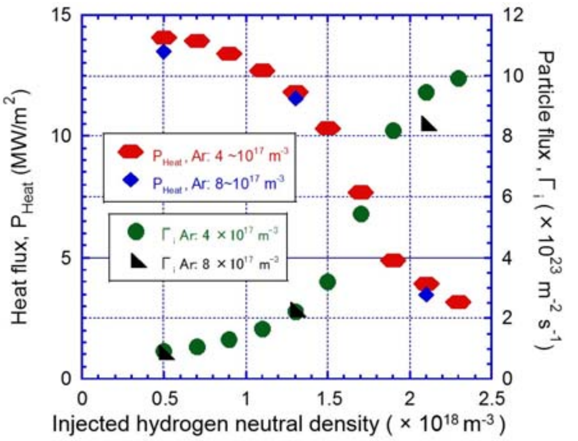

Fig. 4 Dependence of heat load and particle flux on the target plate on injected hydrogen density.

Table 2 The rate of atomic molecular interaction in the energy source term for ion and electron [Ar: 4×10^{17} m^{-3}]

Injected neutral hydrogen [$\times 10^{18}$ m^{-3}]	Ion				Electron			
	Ionization [%]	CX loss between hydrogen neutral and ion [%]	CX between Ar and H^+ [%]	Recombination [%]	Ionization [%]	Radiation cooling [%]	Recombination for Ar^+ [%]	Recombination for H^+ [%]
0.5	8.5	91	6.0×10^{-3}	2.54×10^{-3}	41.7	53.5	2.25×10^{-5}	7.07×10^{-4}
1.7	3.9	96	0.00	0.0124	58.3	41.7	3.65×10^{-5}	6.59×10^{-3}
2.3	2.7	97	0.00	0.0755	57.5	42.3	6.63×10^{-5}	0.045

5 Summary

In this study, the behaviour of the plasma flow at the end-cell of GAMMA 10/PDX has been studied under the various conditions by developing a multi fluid code. We have simulated plasma behaviour from end-cell region to the upstream region (plug/barrier-cell) to evaluate the effect of hydrogen and impurity particles. In this simulation, Ar is considered as impurity particles. In the Ar injection, hydrogen neutral was injected simultaneously to investigate the effect of hydrogen on the plasma parameters. In this condition, Ar neutral density is fixed at 4.0×10^{17} m^{-3} and neutral hydrogen density is changed from 5.0×10^{17} m^{-3} to 2.3×10^{18} m^{-3}. Electron temperature, ion temperature, heat and particle fluxes on the target plate decrease with increasing injected neutral gas density. Under the condition of injecting neutral hydrogen and impurity particles, the charge-exchange loss for ion and radiation loss for electron are strongly enhanced. Therefore, both hydrogen ion temperature and electron temperature decreases to 4 eV and the heat load on the target plate is reduced by 80% compared with the condition of low injection rate of neutral hydrogen and impurity. Particle flux at the target plate increases about 10 times when the injected neutral hydrogen density rises from 5×10^{17} m^{-3} to 2.3×10^{18} m^{-3}. The tendency of saturation in particle flux is observed at higher injected hydrogen neutral density region, which indicates that the end-loss plasma approaches to the plasma detachment state. In the future work, at first we will try to investigate the neutral density profiles on the plasma parameters then we will try to add molecular processes to observe the recombination processes. In addition, we have a plan to consider special boundary conditions, such as achieving divertor box condition (D-module), and introducing neutral distribution more precisely by using the Monte-Carlo code.

Acknowledgements This study was supported by the bidirectional collaboration research program in the University of Tsukuba (NIFS12KUGM066, NIFS11KUGM050). The authors would like to thank the member of the GAMMA 10.

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